

Production of saliva:

salivary glands

- Salivary glands produce 0.5 – 2.0 litres of saliva per day.
- Saliva keeps the oral mucosa moist, contains enzymes (amylase for digesting polysaccharides and lysozyme to destroy the wall of Gram-positive bacteria).
- Beside the three pairs of major salivary glands – the sublingual, submandibular and the parotid glands – minor glands also have to be mentioned.

Minor salivary glands

- Total number of minor salivary glands is estimated around 700-1000. Their diameter is 1-5mm and give approx. 5-8% of total saliva production.
- 3 sets of minor glands in the tongue: anterior or apical lingual gland (of Nuhn-Blandin) - mixed, posterior lingual glands (of Weber) - mucous, glands of the circumvallate and foliate papillae (of von Ebner) - serous.
- Palatine, buccal and labial glands.

Tongue

mucous

serous

Palatine Tonsil

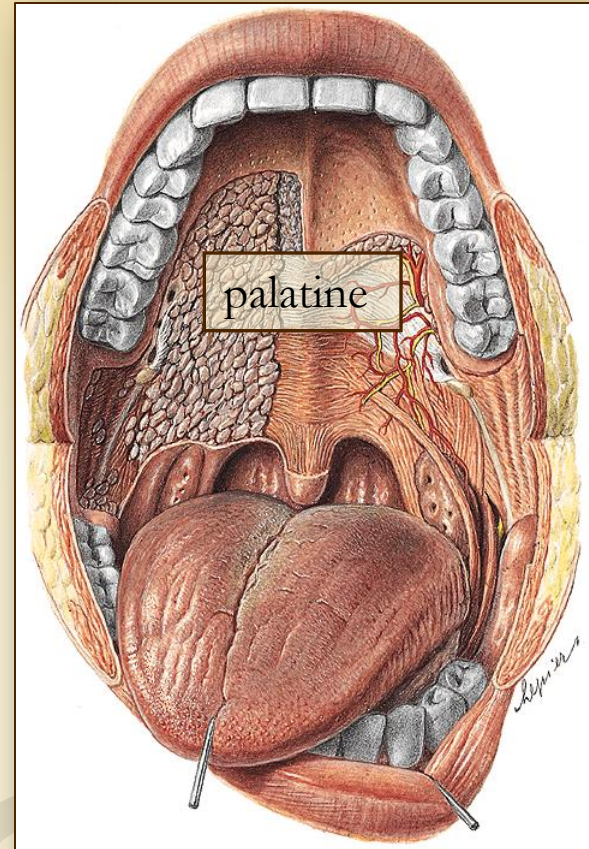
Lingual Tonsil

Circumvallate Papillae

Foliate Papillae

Filiform Papillae

Fungiform Papillae

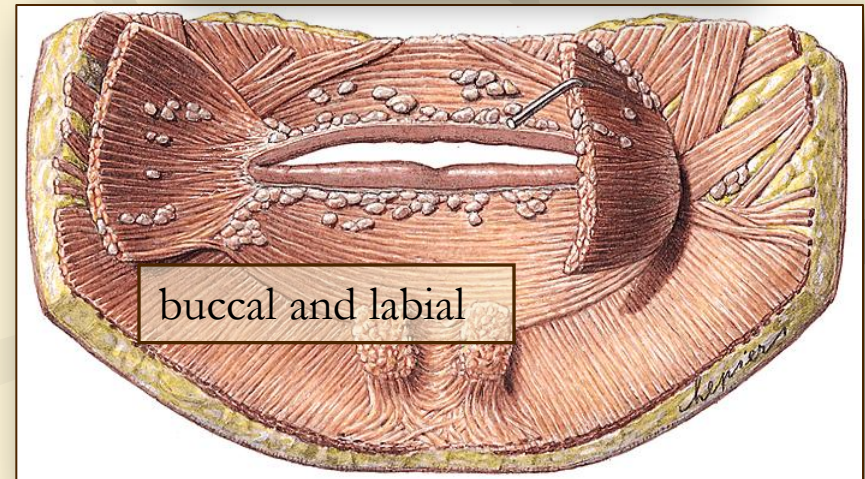
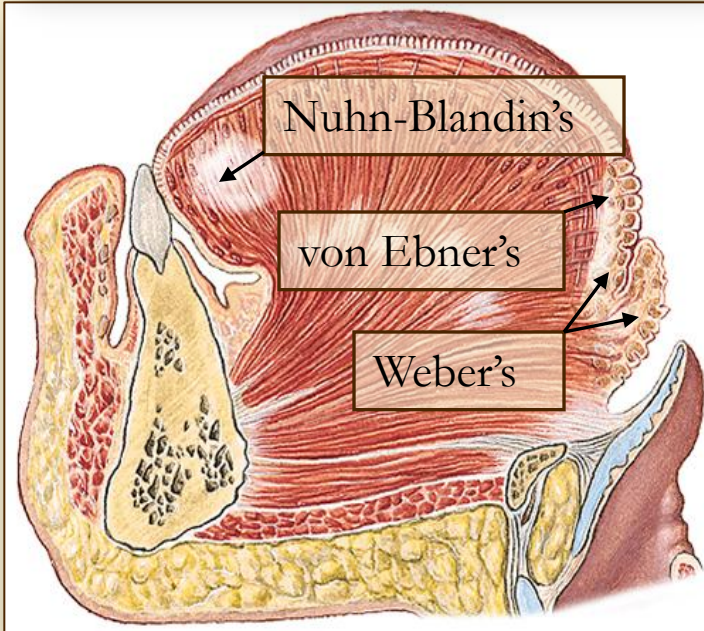


Nuhn-Blandin's

von Ebner's

Weber's

buccal and labial

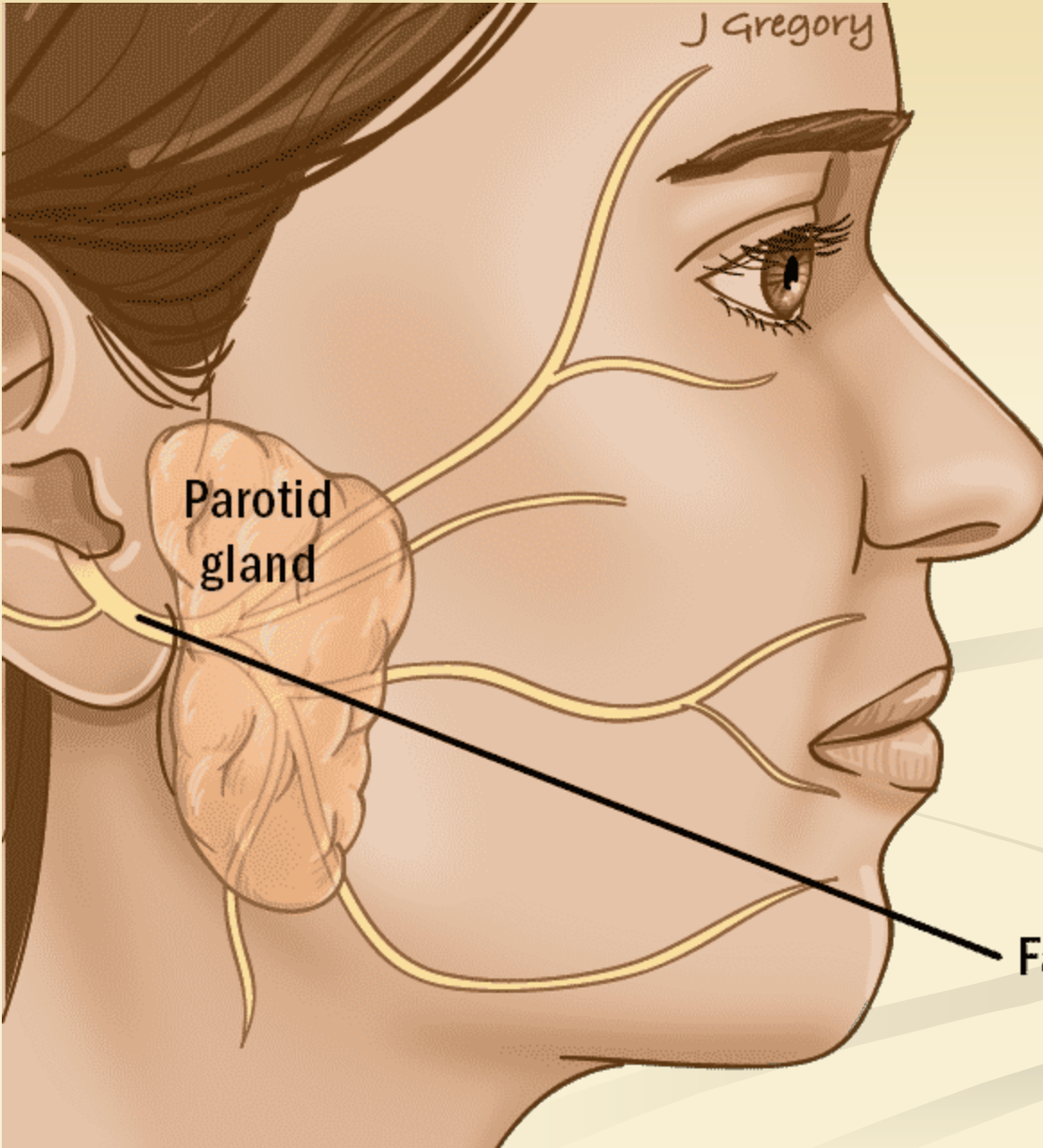


base

apex

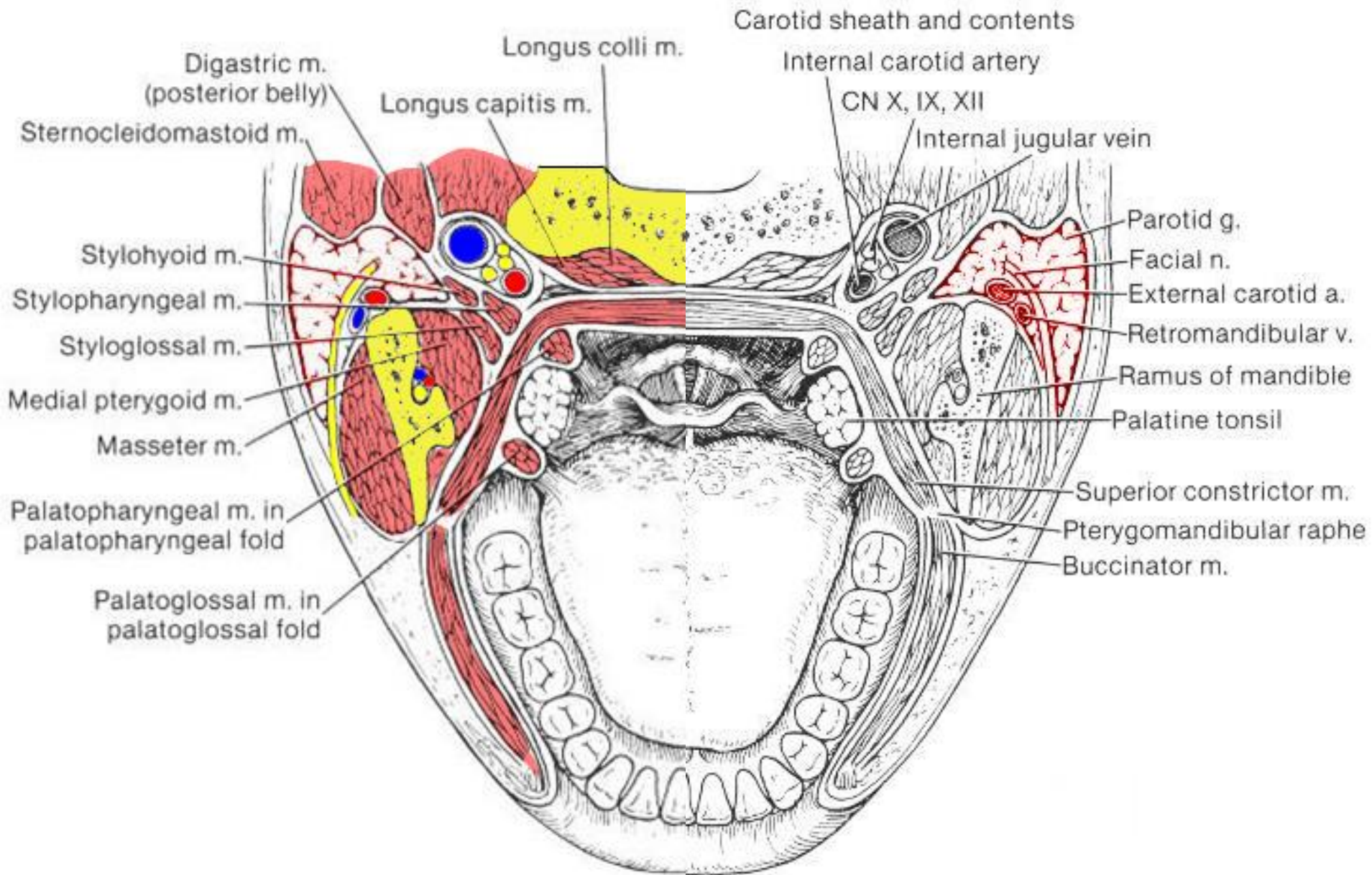
W. H. Jones

Note the accessory parotid gland
joining the parotid duct!



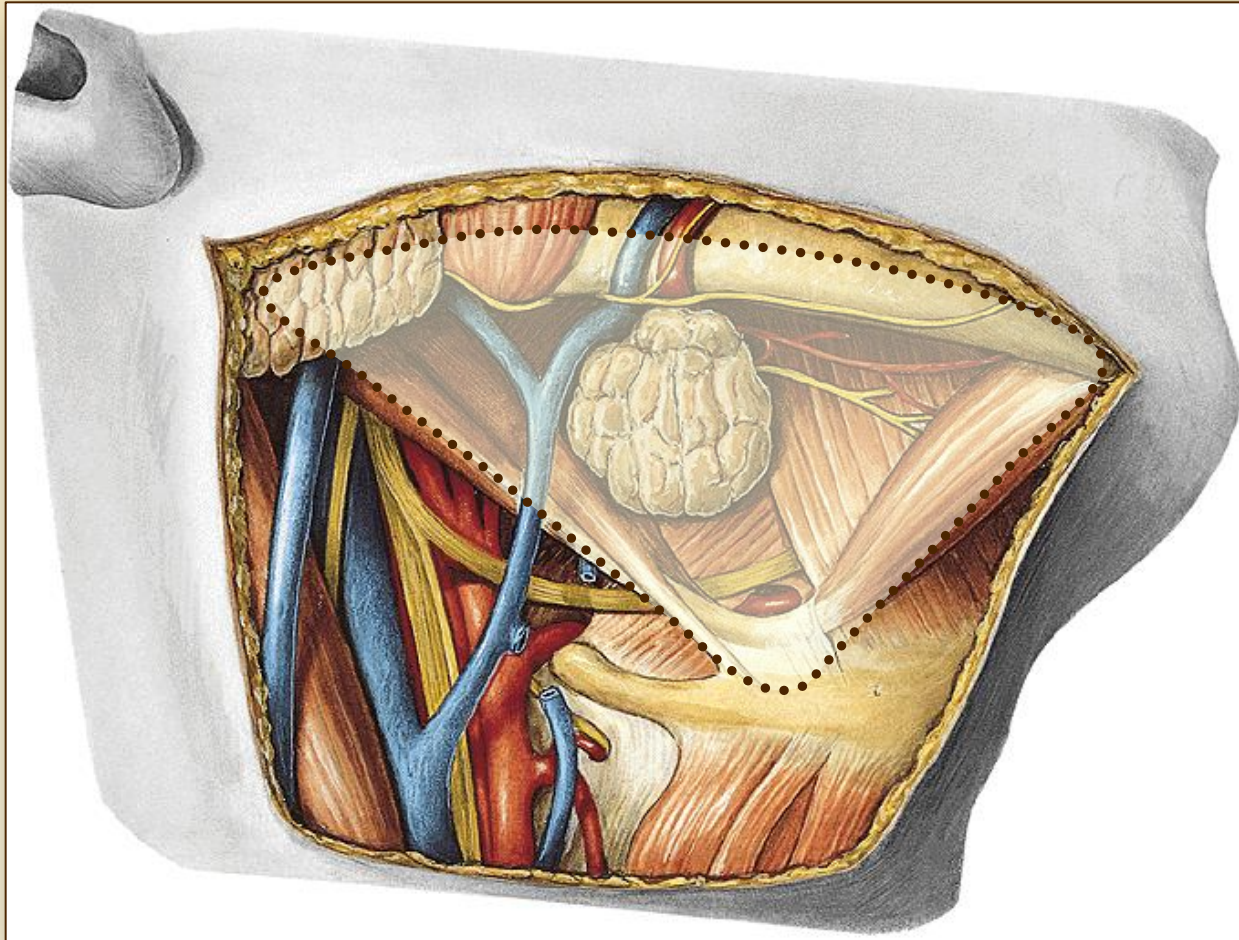
Parotid
gland

Facial nerve



Transverse section demonstrating the **parotid nest**.

Submandibular gland



Submandibular trigone

posterior:

post. belly of digastric
stylohyoid

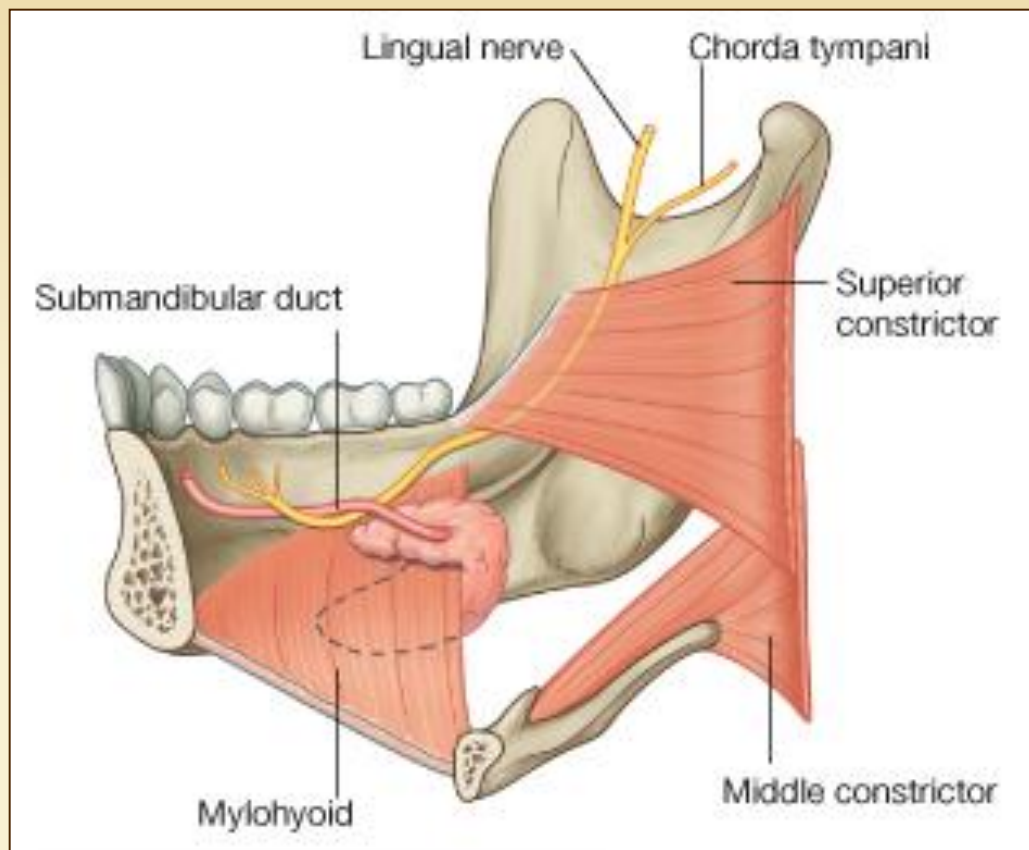
anterior:

ant. belly of digastric

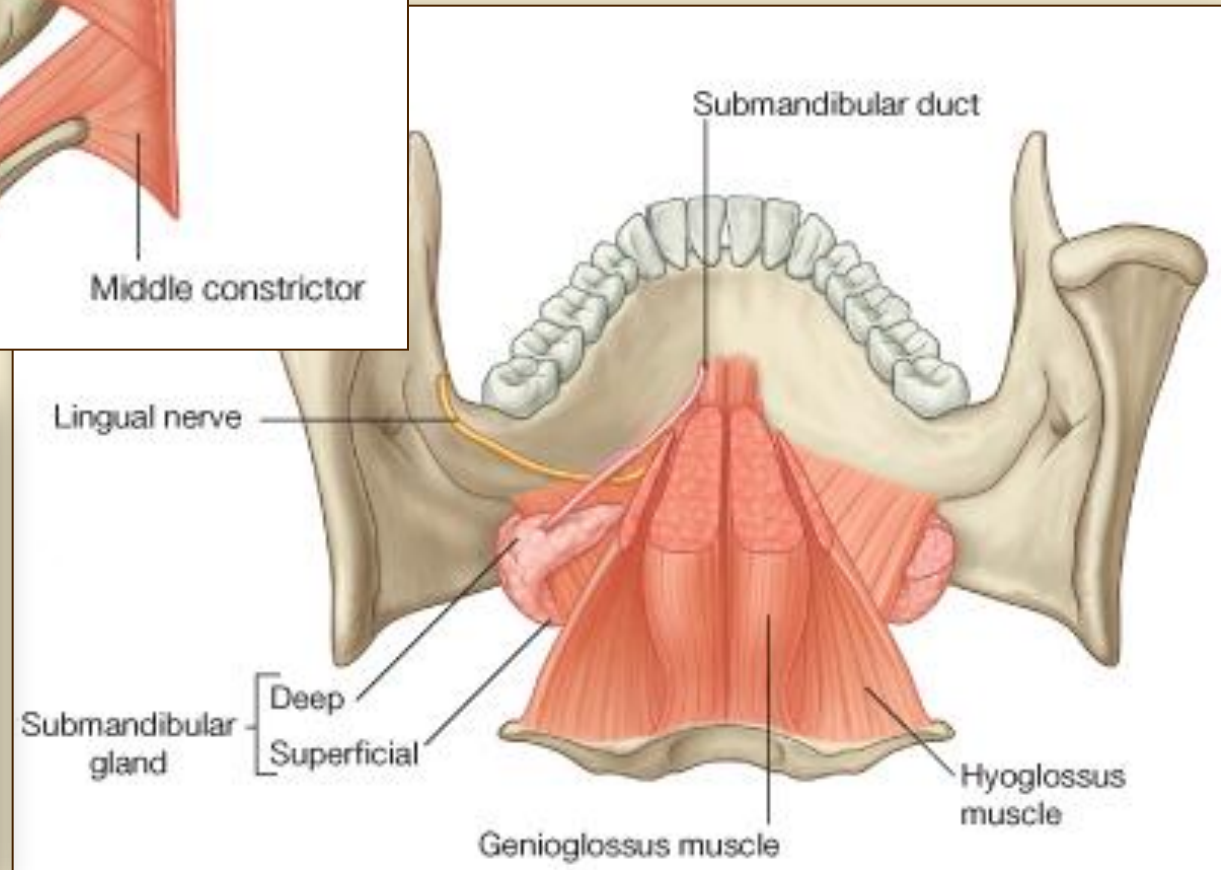
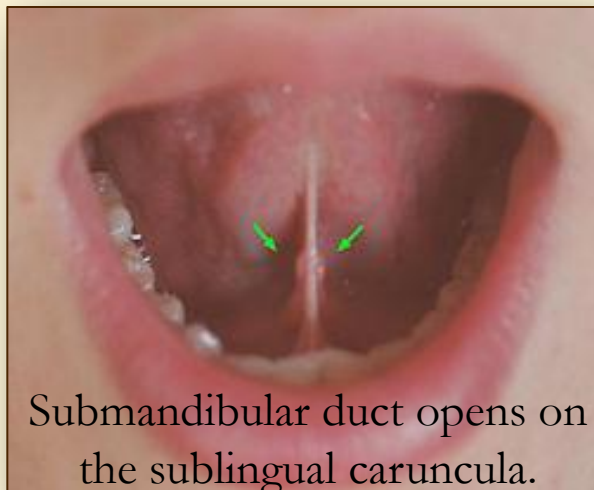
superior:

base of mandible

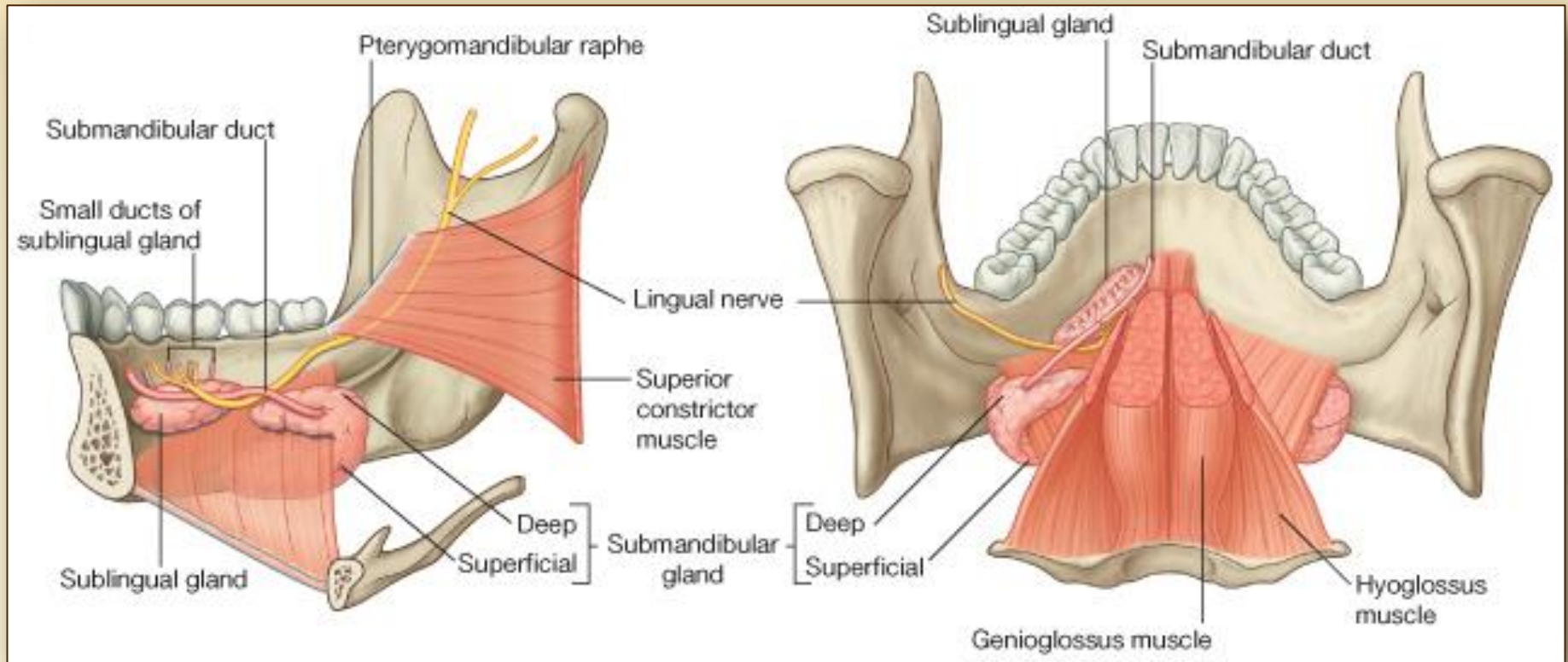
The floor of the trigone, on which the submandibular gland rests, is formed by the mylohyoid and the hyoglossus muscles.



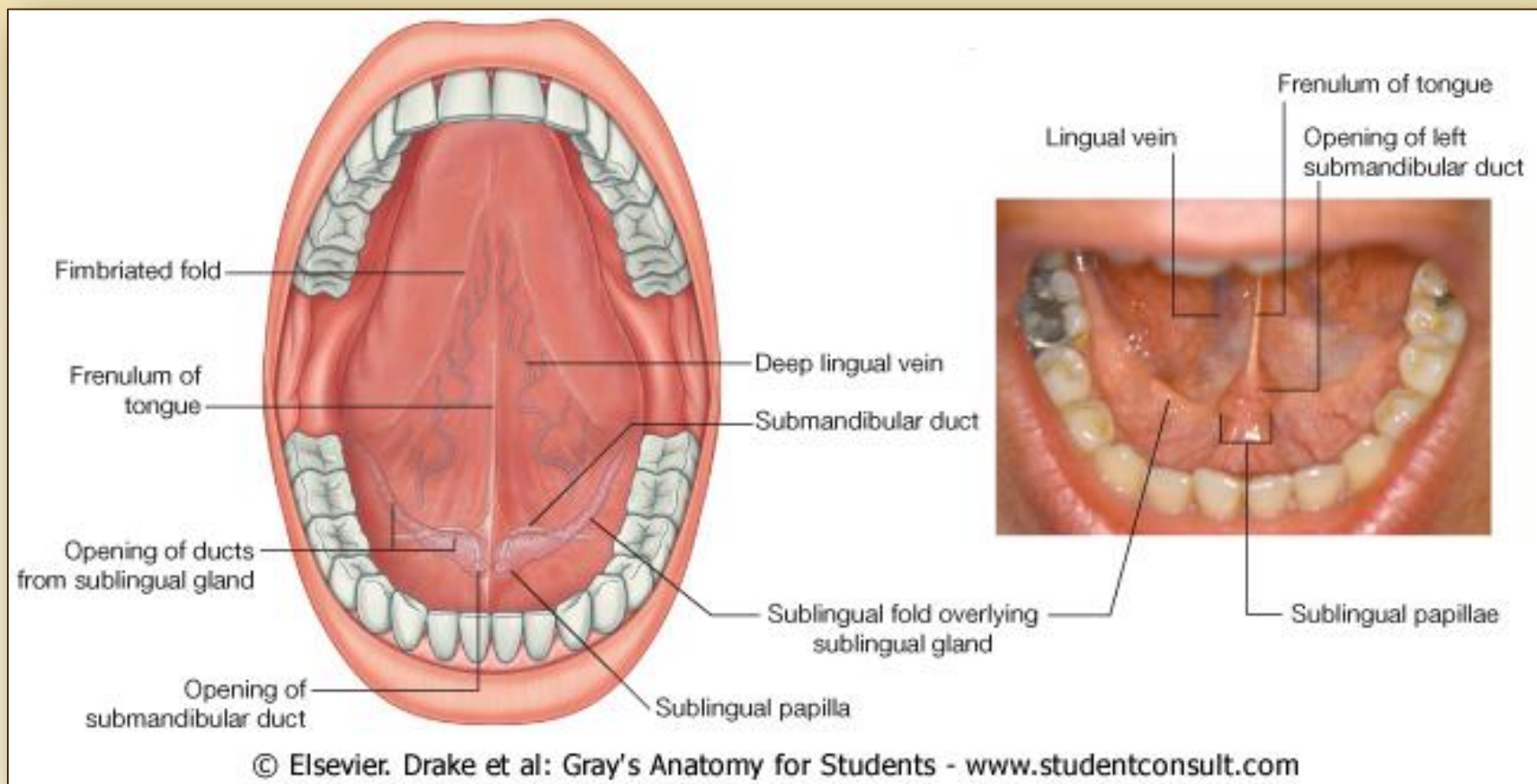
Only the superficial lobe is in the submandibular triangle! Posteriorly the gland invaginates into the lateral lingual sulcus: superficial and deep lobes are separated by the mylohyoid muscle.



Sublingual gland



Sublingual gland rests on the mylohyoid muscle just anterior to the deep lobe of the submandibular gland and is covered by the sublingual mucosa. Laterally is bounded by the mandible (sublingual fossa).



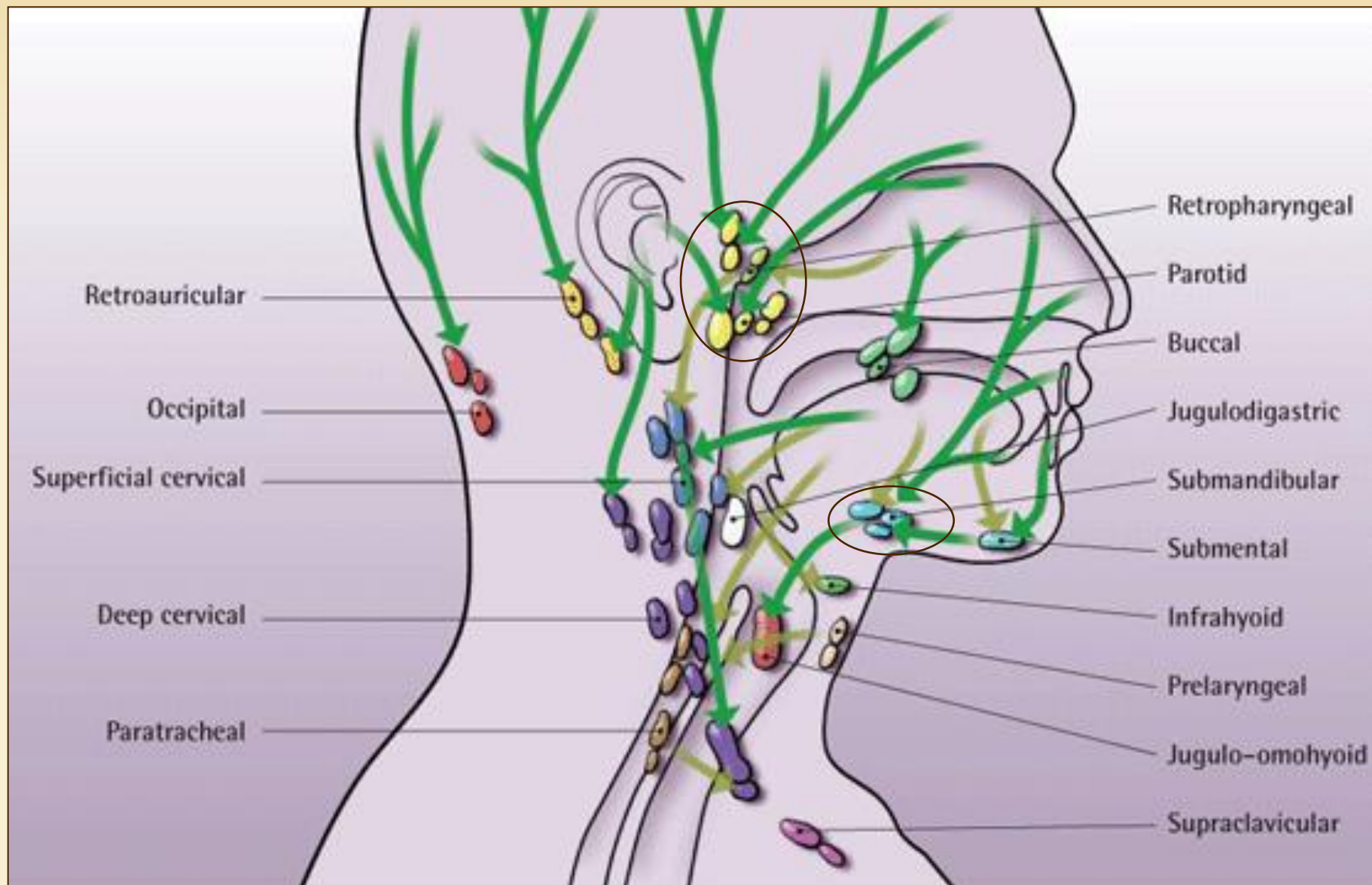
Minor sublingual ducts open along the sublingual fold, while the major sublingual duct unites with the submandibular duct (of Wharton) just before reaching the sublingual caruncula (sublingual papilla).

Blood supply, lymphatic drainage and innervation of salivary glands

Arteries that supply the parotid gland originate from the **external carotid artery** and from its branches that are adjacent to the gland, veins from the parotid gland drain into the **external jugular vein**.

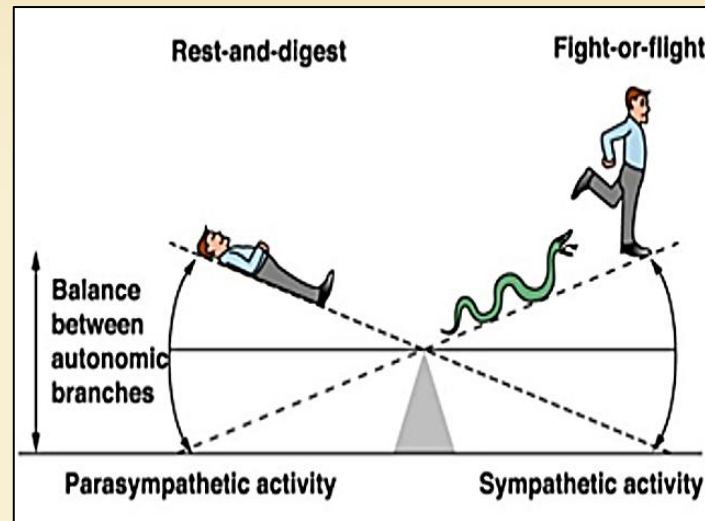
The submandibular and sublingual glands are supplied by branches of the **facial and lingual arteries** and veins from these two glands drain into the **lingual and facial veins**.

Lymphatic vessels from the parotid gland drain into nodes that are on or in the gland. These **parotid nodes** then drain into **superficial and deep cervical nodes**. Lymphatics from the submandibular and sublingual glands drain mainly into **submandibular nodes** and then into deep cervical nodes, particularly the juguloomohyoid node.

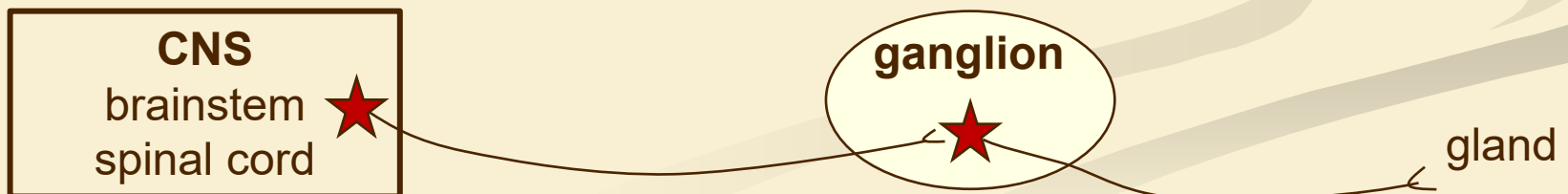


Basic concept:

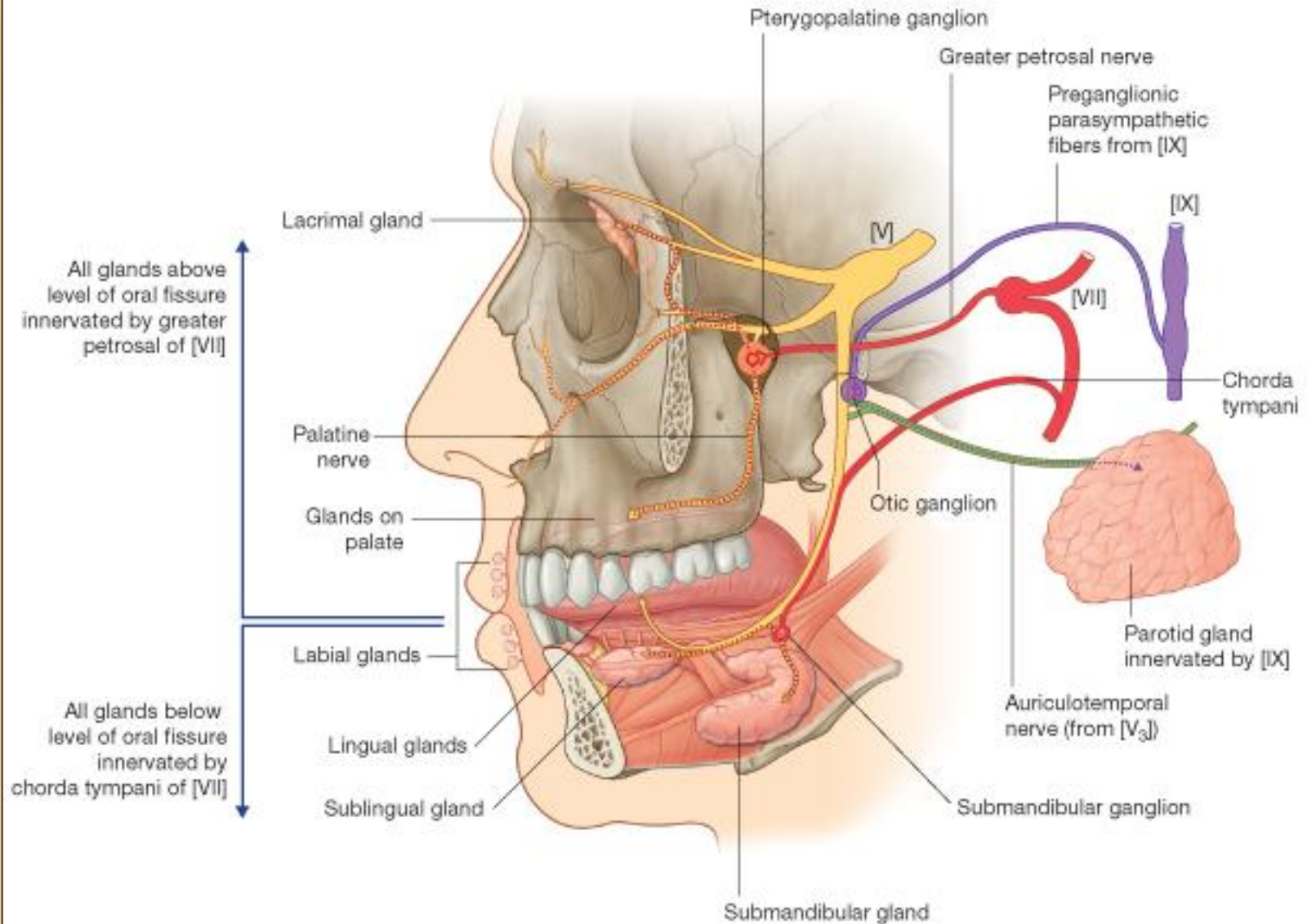
1. Glands are innervated by the autonomic nervous system, usually both sympathetic and parasympathetic fibres terminate in them.



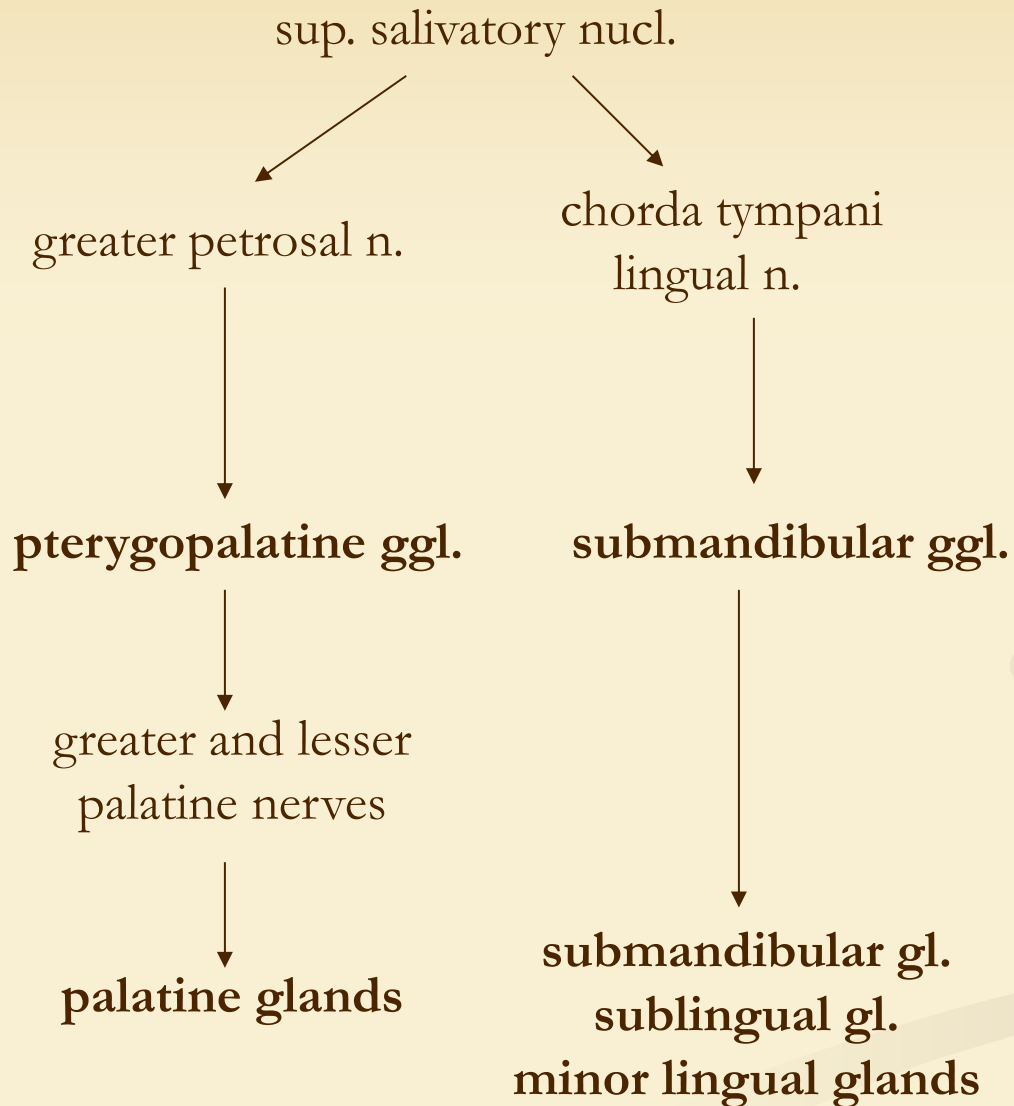
2. CNS does not reach the glands directly: there should be one synapse on the periphery. These synapses are found in autonomic ganglia formed by groups of neurons.



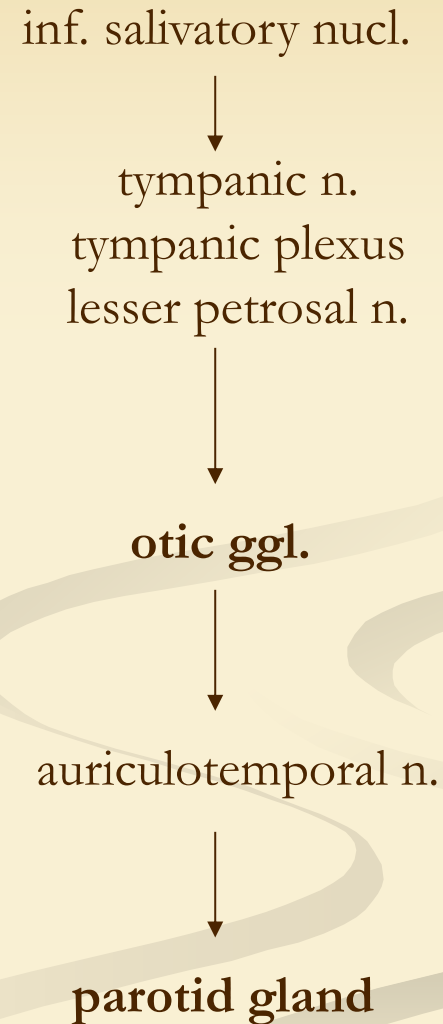
Parasympathetic innervation



Facial nerve



Glossopharyngeal nerve



Sympathetic innervation

Postsynaptic sympathetic fibres arise from the superior cervical ganglion and join the common carotid artery and after its bifurcation the external carotid artery (carotid plexus), and its branches.

